

Inorganic Chemistry Practical: Topic 11 – Group 17 Halides & Precipitation Reactions

10 Most Frequently Asked Practical Examination Questions (Past 4 Years Analysis: 2021–2024)

CANDIDATE NAME	URWAH	CENTRE NUMBER	PK082
COMPONENT	Paper 3 – Advanced Practical Skills (Past 10 Years Archive)	CANDIDATE NUMBER	0142
TIME ALLOWED	2 Hours 30 Minutes	TOTAL MARKS	100 Marks

INSTRUCTIONS TO CANDIDATES:

- Answer all questions in the spaces provided on the question paper.
- Give working for all calculations and record all experimental readings to the required precision:
 - Burette readings to the nearest 0.05 cm^3 ; balance readings to 0.01 g / 0.001 g .
 - Thermometer readings to the nearest $0.5 \text{ }^\circ\text{C}$.
- You may use an HB pencil for any rough working, graphs, or apparatus diagrams.
- Non-exact numerical answers should be given to three significant figures unless stated otherwise.
- You will need a scientific calculator and the Qualitative Analysis Notes provided on pages at the back.

LABORATORY INVESTIGATION OVERVIEW:

This laboratory pack contains authentic Cambridge Advanced Practical examinations. Each practical scenario provides complete procedures, chemical specifications (FA codes), hazard warnings, structured data collection tables, graphical plotting grids, and comprehensive multi-step mathematical and observational deductions.

Question 1: Diagnostic Precipitation of Aqueous Halide Ions with Silver Nitrate**Cambridge Syllabus Reference: 9701/31/M/J/24/Q3 · [10 Marks]**

Three unlabelled solutions, FA 1, FA 2, and FA 3, contain chloride, bromide, and iodide ions. Aqueous silver nitrate is added to each solution after acidification with dilute nitric acid.

(a) Complete the table describing the precipitate color formed with each halide ion. [3]

Halide Ion	Formula of Precipitate	Precipitate Color
Chloride, Cl ⁻	AgCl	White
Bromide, Br ⁻	AgBr	Cream
Iodide, I ⁻	AgI	Yellow

(b) Write the ionic equation with state symbols for the precipitation of silver iodide. [2]

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(c) Explain why dilute nitric acid is added prior to silver nitrate. [3]

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(d) Explain why dilute hydrochloric acid CANNOT be used to acidify the solution. [2]

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Question 2: Differential Solubility of Silver Halide Precipitates in Aqueous Ammonia**Cambridge Syllabus Reference: 9701/33/M/J/23/Q3 · [10 Marks]**

Silver halide precipitates are often difficult to distinguish by visual inspection of color alone. Addition of dilute vs concentrated aqueous ammonia provides unambiguous confirmation.

(a) State the solubility behavior of AgCl, AgBr, and AgI in dilute aqueous ammonia. [3]

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(b) State the solubility behavior of AgBr and AgI in concentrated aqueous ammonia. [2]

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(c) Explain why AgCl dissolves in dilute NH₃ while AgI is insoluble even in concentrated NH₃. [3]

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(d) Write the formula of the complex ion formed when AgCl dissolves in ammonia. [2]

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Question 3: Halogen Displacement Reactions and Non-Polar Extraction into Cyclohexane

Cambridge Syllabus Reference: 9701/32/M/J/23/Q3 · [10 Marks]

Aqueous chlorine water is added to separate solutions of NaBr(aq) and NaI(aq). Cyclohexane (density = 0.78 g cm^{-3}) is added, and the test-tubes are stoppered and shaken vigorously.

(a) State the color observed in the upper organic (cyclohexane) layer for the NaBr test. [2]

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(b) State the color observed in the upper organic layer for the NaI test. [2]

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(c) Explain why halogens dissolve preferentially in cyclohexane rather than water. [3]

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(d) Write the ionic equation for the displacement of iodide ions by chlorine. [3]

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Question 4: Reactions of Solid Sodium Chloride and Sodium Bromide with Concentrated Sulfuric Acid

Cambridge Syllabus Reference: 9701/34/O/N/23/Q3 · [10 Marks]

Solid samples of NaCl and NaBr are placed in dry boiling tubes in a fume cupboard. Concentrated sulfuric acid is added dropwise to each tube.

(a) State the observations when concentrated H₂SO₄ is added to solid NaCl. [2]

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(b) State the observations when concentrated H₂SO₄ is added to solid NaBr. [3]

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(c) Identify the type of reaction occurring with NaCl and explain why no redox occurs. [2]

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(d) Write the balanced redox equation for the oxidation of bromide ions by concentrated sulfuric acid. [3]

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Question 5: Reactions of Solid Sodium Iodide with Concentrated Sulfuric Acid: Deep Reduction

Cambridge Syllabus Reference: 9701/31/O/N/22/Q3 · [10 Marks]

When concentrated sulfuric acid is added to solid sodium iodide, an exothermic reaction occurs. Multiple products are detected, including purple vapors, a dark solid, a yellow deposit, and an offensive odor.

(a) Identify the chemicals responsible for: (i) purple vapor, (ii) yellow solid, (iii) bad egg smell. [3]

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(b) State the changes in oxidation number of sulfur in H_2SO_4 when it is reduced to: (i) SO_2 , (ii) S, (iii) H_2S . [3]

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(c) Write the half-equation for the reduction of H_2SO_4 to H_2S . [2]

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(d) Write the overall balanced equation for the reaction producing H_2S and I_2 . [2]

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Question 6: Theoretical Justification of Reducing Ability Trends of Halide Ions Down Group 17

Cambridge Syllabus Reference: 9701/35/M/J/24/Q3 · [10 Marks]

The experimental reactions with concentrated sulfuric acid demonstrate that the reducing strength of halide ions increases: $\text{Cl}^- < \text{Br}^- < \text{I}^-$.

(a) Define 'reducing agent' in terms of electron transfer. [2]

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(b) Explain in terms of atomic structure why iodide is a much stronger reducing agent than chloride. [4]

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(c) Predict whether fluoride, F^- , would react with concentrated H_2SO_4 in a redox reaction. Justify. [2]

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(d) State the formula of the dangerous gas produced when NaF reacts with conc H_2SO_4 . [2]

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Question 7: Analytical Methodology: Rationale for Nitric Acid Addition in Halide Analysis

Cambridge Syllabus Reference: 9701/32/M/J/24/Q3 · [10 Marks]

A student tests an unknown solution X by adding silver nitrate directly without adding dilute nitric acid first. A pale yellow-white precipitate forms. The student concludes that solution X contains bromide ions. In reality, solution X contains sodium carbonate, Na_2CO_3 .

(a) Identify the precipitate formed when AgNO_3 is added to aqueous Na_2CO_3 . [2]

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(b) Explain how adding dilute HNO_3 prevents this false positive conclusion. [3]

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(c) Explain why sulfuric acid, H_2SO_4 , cannot be used to acidify before AgNO_3 . [3]

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(d) Write the balanced ionic equation for the reaction of Ag_2CO_3 with nitric acid. [2]

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Question 8: Coordination Chemistry: Dissolution of Silver Chloride in Aqueous Ammonia

Cambridge Syllabus Reference: 9701/33/O/N/22/Q3 · [10 Marks]

When excess dilute aqueous ammonia is added to a test-tube containing a precipitate of silver chloride, the cloudy suspension dissolves completely to form a transparent colorless solution.

(a) Write the balanced chemical equation for the dissolution of AgCl in aqueous ammonia. [3]

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(b) State the coordination number and the shape of the $[\text{Ag}(\text{NH}_3)_2]^+$ complex ion. [2]

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(c) Identify the type of bond between the silver ion and the nitrogen atom of ammonia. [2]

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(d) Describe what happens when dilute nitric acid is added to the resulting clear solution. [3]

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Question 9: Discrimination Between Chloride (Cl^-) and Chlorate(I) (ClO^-) in Commercial Bleach

Cambridge Syllabus Reference: 9701/34/M/J/22/Q3 · [10 Marks]

Household bleach contains both sodium chlorate(I), NaClO , and sodium chloride, NaCl .
A student must design tests to distinguish between Cl^- and ClO^- ions in the solution.

(a) State the oxidation state of chlorine in: (i) Cl^- , (ii) ClO^- . [2]

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(b) Describe a chemical test that demonstrates the bleaching (oxidizing) action of ClO^- . [3]

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(c) When dilute sulfuric acid is added to bleach, chlorine gas is evolved. Write the ionic equation. [3]

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(d) Identify the type of redox reaction described in (c). [2]

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Question 10: Photochemical Decomposition of Silver Halides and Storage in Amber Glassware

Cambridge Syllabus Reference: 9701/35/O/N/23/Q3 · [10 Marks]

When freshly precipitated white AgCl is left on a watch-glass exposed to bright laboratory light, it gradually darkens through lavender to grey-violet and finally dark grey/black.

(a) Write the balanced chemical equation for the photochemical decomposition of silver chloride. [3]

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(b) Identify the grey-black substance formed on the surface of the precipitate. [2]

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(c) Explain why aqueous silver nitrate must be stored in dark brown (amber) glass bottles. [2]

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(d) State the historical industrial application of this photochemical property. [3]

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Examiner Mark Scheme & Indicative Observation Guide

Confidential Marking Points & Error Tolerances for Inorganic Chemistry Practical: Topic 11 – Group 17 Halides & Precipitation Reactions

Question 1: Diagnostic Precipitation of Aqueous Halide Ions with Silver Nitrate [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	AgCl = White [1]; AgBr = Cream [1]; AgI = Yellow [1].	[3]
(b)	$\text{Ag}^+(\text{aq}) + \text{I}^-(\text{aq}) \rightarrow \text{AgI}(\text{s})$ [2].	[2]
(c)	To decompose any carbonate (CO_3^{2-}) or sulfite (SO_3^{2-}) ions [1.5], preventing the formation of insoluble Ag_2CO_3 or Ag_2SO_3 which would give a false positive precipitate [1.5].	[3]
(d)	HCl contains Cl^- ions which would immediately precipitate white AgCl with silver nitrate, invalidating the test [2].	[2]

Question 2: Differential Solubility of Silver Halide Precipitates in Aqueous Ammonia [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	AgCl: Dissolves completely to form a colorless solution [1]; AgBr: Insoluble / does not dissolve in dilute NH_3 [1]; AgI: Insoluble in dilute NH_3 [1].	[3]
(b)	AgBr: Dissolves completely in concentrated NH_3 [1]; AgI: Remains insoluble in concentrated NH_3 [1].	[2]
(c)	AgCl has a higher solubility product (K_{sp}) than AgI [1]; complexing of Ag^+ with NH_3 lowers $[\text{Ag}^+]$ enough to exceed the K_{sp} threshold for AgCl [1]; for AgI, K_{sp} is so extremely small that even with concentrated NH_3 $[\text{Ag}^+][\text{I}^-]$ remains greater than K_{sp} [1].	[3]
(d)	$[\text{Ag}(\text{NH}_3)_2]^+$ (diamminesilver(I) ion) [2].	[2]

Question 3: Halogen Displacement Reactions and Non-Polar Extraction into Cyclohexane [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Orange / reddish-brown (due to dissolved non-polar bromine, Br_2) [2].	[2]
(b)	Bright violet / purple (due to dissolved non-polar iodine, I_2) [2].	[2]
(c)	Halogen molecules (Br_2 , I_2) are non-polar [1]; they form favorable London dispersion forces with non-polar cyclohexane molecules [1], whereas they cannot form hydrogen bonds with polar water molecules [1].	[3]
(d)	$\text{Cl}_2(\text{aq}) + 2\text{I}^-(\text{aq}) \rightarrow 2\text{Cl}^-(\text{aq}) + \text{I}_2(\text{aq})$ [3] (1 mark for species, 1 mark for balancing, 1 mark for state symbols).	[3]

Question 4: Reactions of Solid Sodium Chloride and Sodium Bromide with Concentrated Sulfuric Acid [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Steamy / misty acidic fumes of HCl gas [1]; effervescence / solid dissolves [1].	[2]
(b)	Steamy fumes of HBr [1]; reddish-brown / orange vapor of Br_2 gas [1]; choking pungent gas (SO_2) evolved [1].	[3]
(c)	Acid-base (proton transfer) reaction [1]; chloride (Cl^-) is too weak a reducing agent to reduce concentrated H_2SO_4 [1].	[2]
(d)	$2\text{H}^+ + \text{H}_2\text{SO}_4 + 2\text{Br}^- \rightarrow \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$ (or $2\text{NaBr} + 2\text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + \text{Br}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$) [3].	[3]

Question 5: Reactions of Solid Sodium Iodide with Concentrated Sulfuric Acid: Deep Reduction [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	(i) Purple vapor: Iodine (I_2) [1]; (ii) Yellow solid: Sulfur (S) [1]; (iii) Bad egg smell: Hydrogen sulfide (H_2S) [1].	[3]

(b)	In H ₂ SO ₄ , S is +6; (i) In SO ₂ : +4 (reduction of 2) [1]; (ii) In S: 0 (reduction of 6) [1]; (iii) In H ₂ S: -2 (reduction of 8) [1].	[3]
(c)	H ₂ SO ₄ + 8H ⁺ + 8e ⁻ → H ₂ S + 4H ₂ O (or SO ₄ ²⁻ + 10H ⁺ + 8e ⁻ → H ₂ S + 4H ₂ O) [2].	[2]
(d)	8NaI + 5H ₂ SO ₄ → 4Na ₂ SO ₄ + 4I ₂ + H ₂ S + 4H ₂ O (or 8I ⁻ + H ₂ SO ₄ + 8H ⁺ → 4I ₂ + H ₂ S + 4H ₂ O) [2].	[2]

Question 6: Theoretical Justification of Reducing Ability Trends of Halide Ions Down Group 17 [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	An electron donor / species that loses electrons to reduce another substance [2].	[2]
(b)	Down Group 17, ionic radius increases significantly (I ⁻ > Br ⁻ > Cl ⁻) [1]; valence electrons are in higher energy levels with greater distance from nucleus [1]; increased shielding by inner electron shells weakens electrostatic attraction between nucleus and outer electrons [1]; outer electrons are lost much more readily [1].	[4]
(c)	No redox reaction occurs [1]; fluoride is an even weaker reducing agent than chloride due to very small ionic radius and extremely tight electron hold [1].	[2]
(d)	Hydrogen fluoride, HF (highly corrosive gas that etches glass) [2].	[2]

Question 7: Analytical Methodology: Rationale for Nitric Acid Addition in Halide Analysis [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	Silver carbonate, Ag ₂ CO ₃ (pale yellow / off-white precipitate) [2].	[2]
(b)	Nitric acid reacts with carbonate ions to form soluble nitrate, CO ₂ gas, and water: 2H ⁺ + CO ₃ ²⁻ → H ₂ O + CO ₂ [2]; this completely eliminates carbonate ions before silver ions are introduced [1].	[3]
(c)	Sulfuric acid contains sulfate ions (SO ₄ ²⁻) [1]; silver sulfate (Ag ₂ SO ₄) is sparingly soluble and can precipitate as a white solid [1], which would interfere with the halide test [1].	[3]
(d)	Ag ₂ CO ₃ (s) + 2H ⁺ (aq) → 2Ag ⁺ (aq) + CO ₂ (g) + H ₂ O(l) [2].	[2]

Question 8: Coordination Chemistry: Dissolution of Silver Chloride in Aqueous Ammonia [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	AgCl(s) + 2NH ₃ (aq) → [Ag(NH ₃) ₂] ⁺ (aq) + Cl ⁻ (aq) [3] (or AgCl + 2NH ₃ → [Ag(NH ₃) ₂]Cl).	[3]
(b)	Coordination number: 2 [1]; Shape: Linear (180° bond angle) [1].	[2]
(c)	Dative covalent (coordinate) bond [2].	[2]
(d)	Dilute HNO ₃ neutralizes the NH ₃ ligands to form NH ₄ ⁺ [1.5]; the diamminesilver complex is destroyed and white AgCl precipitates again [1.5].	[3]

Question 9: Discrimination Between Chloride (Cl⁻) and Chlorate(I) (ClO⁻) in Commercial Bleach [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	(i) In Cl ⁻ : -1 [1]; (ii) In ClO ⁻ : +1 [1].	[2]
(b)	Add blue litmus paper: the litmus turns red momentarily due to acidity and then is rapidly bleached completely white [3].	[3]
(c)	ClO ⁻ (aq) + Cl ⁻ (aq) + 2H ⁺ (aq) → Cl ₂ (g) + H ₂ O(l) [3].	[3]
(d)	Comproportionation (synproportionation) [2].	[2]

Question 10: Photochemical Decomposition of Silver Halides and Storage in Amber Glassware [10 Marks]

Part	Marking Points & Expected Observations	Marks
(a)	2AgCl(s) → 2Ag(s) + Cl ₂ (g) [3] (under UV / visible light).	[3]
(b)	Finely divided metallic silver, Ag(s) [2].	[2]

(c)	Amber glass filters out UV and high-energy visible light, preventing photochemical reduction of Ag^+ to Ag metal during storage [2].	[2]
(d)	Traditional black-and-white photographic film and photographic paper development [3].	[3]